

1. Explain the difference between `map()`, `filter()`, and `reduce()` in Python.

- What does each function return?
- When should you prefer lambda functions over normal functions?

2. Given two lists:

```
a = [1, 2, 3, 4] b = [10, 20, 30, 40]
```

Use `map()` with a lambda to create a new list containing the sum of corresponding elements.

What happens if the lists are of unequal length?

3. Given a list:

```
nums = [12, 15, 7, 18, 20, 21, 25]
```

Use `filter()` and lambda to keep numbers that are divisible by 3 OR divisible by 5 but NOT divisible by both.

Explain how the logical condition works.

4. Given a list:

```
nums = [1, 2, 3, 4]
```

Use `reduce()` with a lambda to compute the sum, but start with an initial value of 10.

Explain how the initial value affects the reduction process.

5. Consider the code below:

```
nums = [[1, 2], [3, 4], [5, 6]] result = list(map(lambda x: x.append(10), nums))  
print("Result:", result) print("Nums:", nums)
```

Questions

- What will be the output of `result`?
- What will be the output of `nums`?
- Why does `map()` behave this way with `list.append()`?
- How can you modify the lambda so that `nums` is not changed?